Predicting Traffic Accident Severity Using Machine Learning

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Business Problem

Road traffic accidents continue to result in significant loss of life, injuries, and economic damage despite ongoing improvements in vehicle safety technologies and traffic management systems. Advanced Driver Assistance Systems (ADAS) and emerging autonomous vehicle technologies rely heavily on real-time risk assessment to support safer driving decisions. However, many existing systems focus on immediate hazard detection rather than predicting the potential severity of an incident.

The objective of this study is to develop a machine learning framework capable of predicting traffic accident severity using historical crash data and environmental conditions. By identifying patterns associated with severe crashes, the results can support vehicle manufacturers, transportation planners, and safety engineers in designing proactive safety features and risk-aware driving systems. The project also aims to determine which environmental, temporal, and roadway factors contribute most significantly to high-severity accidents.

Background and History

Traffic safety has been a major public health concern in the United States for decades. According to the National Highway Traffic Safety Administration (NHTSA), thousands of fatalities and millions of injuries occur each year due to roadway incidents. With the growth of connected vehicles, sensor networks, and large-scale transportation datasets, data-driven safety analysis has become an essential component of modern transportation research.

Recent studies have demonstrated that machine learning techniques can identify complex relationships between crash outcomes and contributing factors such as weather, lighting conditions, traffic control infrastructure, and time of day. The US Accidents dataset compiled by

Moosavi et al. provides a large-scale, multi-source collection of traffic incident records across the country, making it suitable for predictive modeling and safety analysis. Leveraging such datasets allows researchers to move beyond descriptive analysis toward predictive and prescriptive insights that support intelligent transportation systems.

Data Explanation and Preparation

This study utilizes the US_Accidents_March23 dataset obtained from Kaggle, which contains millions of accident records collected from traffic sensors, public agencies, and incident reports across the United States. The dataset includes information related to accident severity, geographic location, time, weather conditions, visibility, roadway characteristics, and nearby infrastructure features.

Data preparation involved several preprocessing steps to ensure quality and usability for modeling. Records with missing or inconsistent severity labels were removed. Temporal fields such as start time were converted into structured features including hour of day, day of week, and month to capture temporal patterns. Weather-related variables such as temperature, humidity, visibility, wind speed, and precipitation were cleaned and standardized.

Categorical variables including weather condition, wind direction, and daylight status were encoded for machine learning compatibility. Boolean roadway indicators such as the presence of traffic signals, junctions, crossings, and railway intersections were retained as binary features. Since the severity variable is ordinal and imbalanced across categories, class distribution was analyzed and resampling techniques were considered to improve model performance on severe cases.

Exploratory Analysis and Modeling Results

The exploratory analysis and modeling results are presented in Figures 1 through 6. These visualizations illustrate the distribution of accident severity, temporal patterns, environmental influences, model performance, key predictive factors, and classification behavior.

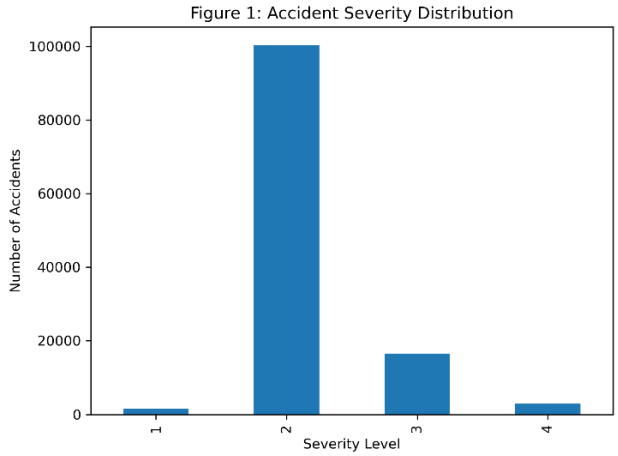


Figure 1. *Distribution of traffic accident severity levels showing class imbalance.*

Figure 1 shows the distribution of accident severity levels. The results indicate a clear class imbalance, with most records concentrated on lower severity categories. This imbalance is important because predictive models may become biased toward the dominant classes, making accurate identification of high-severity accidents more challenging.

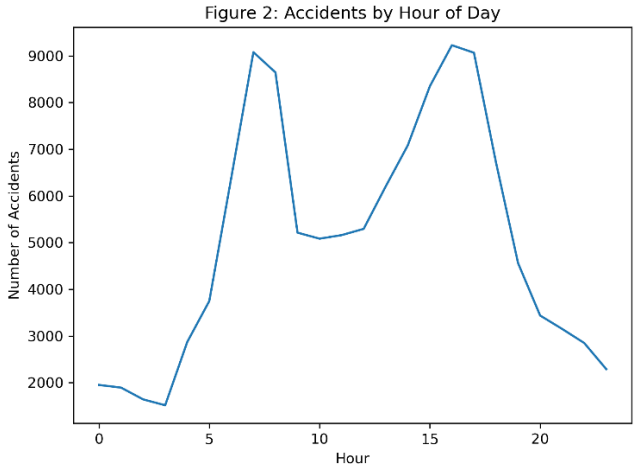


Figure 2. *Number of accidents by hour of day, illustrating temporal patterns in crash occurrence.*

Figure 2 demonstrates that accident frequency varies throughout the day, with peaks observed during early morning and late-night hours, likely due to reduced visibility and driver fatigue.

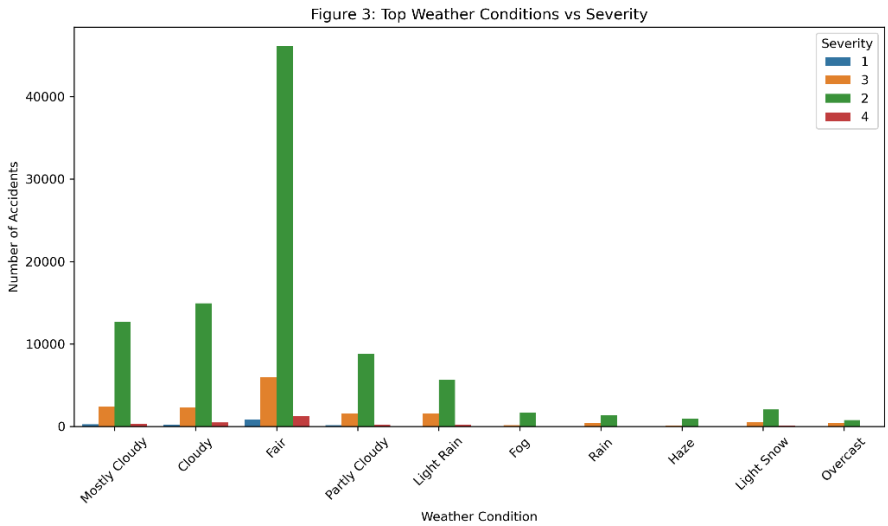


Figure 3. *Relationship between major weather conditions and accident severity highlights the impact of environmental factors.*

Figure 3 shows that adverse weather conditions such as rain, fog, and low visibility are associated with increased accident severity.

Table 1 presents evaluation metrics for Logistic Regression, Random Forest, and Gradient Boosting models. Logistic Regression and Gradient Boosting achieved the highest overall performance (~82% accuracy), while Random Forest performed slightly lower. Weighted precision, recall, and F1-scores account for the class imbalance in accident severity levels.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8265	0.6832	0.8265	0.7481
Random Forest	0.8131	0.7210	0.8131	0.7505
Gradient Boosting	0.8264	0.7211	0.8264	0.7485

Table 1. *Evaluation metrics for Logistic Regression, Random Forest, and Gradient Boosting models.*

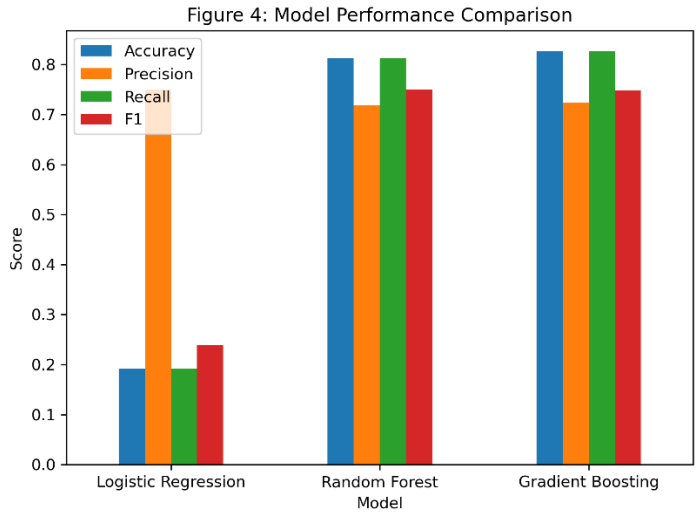


Figure 4. *Performance comparison of Logistic Regression, Random Forest, and Gradient Boosting models based on accuracy, precision, recall, and F1-score.*

Figure 4 compares the performance of the three models based on accuracy, precision, recall, and F1-score. Ensemble-based methods provide slightly better predictive stability, while Logistic Regression offers comparable performance with lower computational complexity.

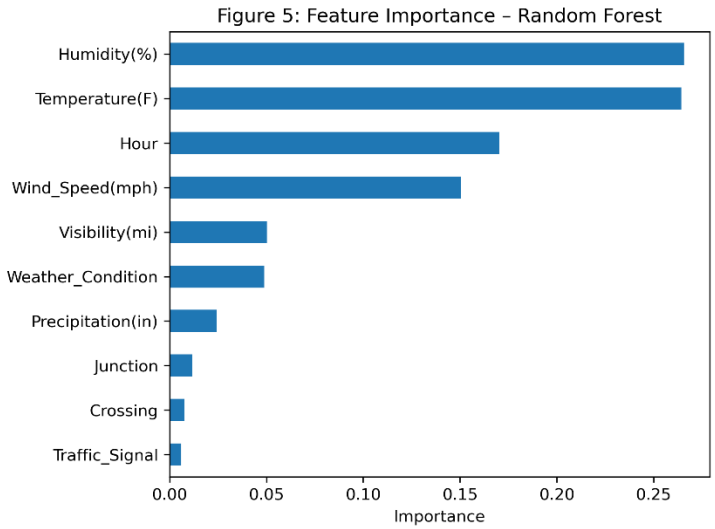


Figure 5. *Feature importance analysis from the Random Forest model identifying the most influential predictors of accident severity.*

Figure 5 shows feature importance derived from the Random Forest model. Environmental variables such as visibility and weather conditions, temporal factors including time of day, and roadway features like the presence of traffic signals were identified as the most influential predictors. These insights provide actionable guidance for ADAS and intelligent transportation applications.

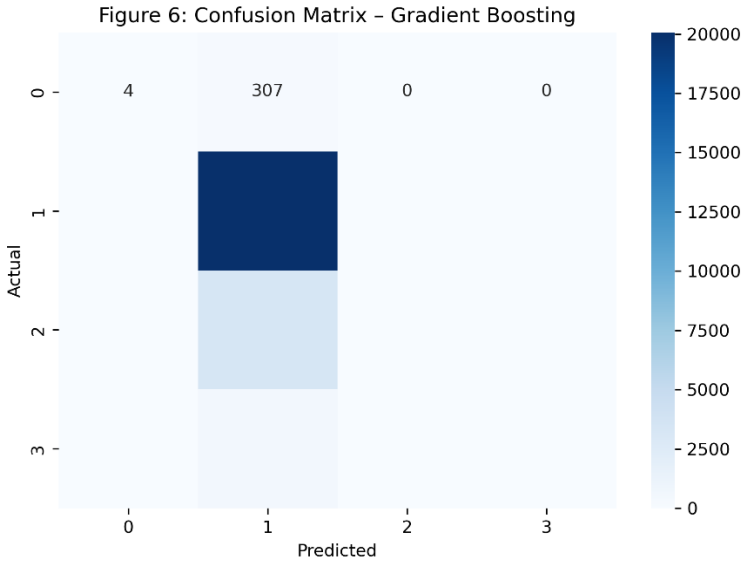


Figure 6. *Confusion matrix for the Gradient Boosting model showing prediction accuracy across accident severity levels.*

Figure 6 presents the confusion matrix for the Gradient Boosting model. While the models perform well overall, predicting the most severe accident categories remains challenging due to their lower frequency.

Methods

The project follows an end-to-end machine learning workflow. Exploratory data analysis was conducted to understand the distribution of accident severity across time, weather conditions, and roadway environments. Correlation analysis and visualization techniques were used to identify potential predictors of severe crashes.

Three supervised learning algorithms were developed and compared: Logistic Regression as a baseline model, Random Forest as an ensemble tree-based method, and Gradient Boosting for improved predictive performance. The dataset was split into training and testing subsets to evaluate generalization capability. Model performance was assessed using accuracy, precision, recall, and F1-score, with particular emphasis on recall for higher severity levels, as failing to identify severe incidents represents a greater safety risk than false alarms. Receiver Operating Characteristic (ROC) analysis was also used to evaluate classification performance. Feature importance techniques identified the most influential variables affecting severity predictions.

Analysis

Preliminary analysis revealed that accident severity is influenced by a combination of environmental, temporal, and infrastructure-related factors. Severe accidents were more likely to occur during low-visibility conditions such as night or adverse weather. Weather conditions including rain, fog, and reduced visibility showed a measurable association with higher severity levels.

Temporal patterns indicated increased severity during early morning and late-night hours. Infrastructure features such as the absence of traffic control devices and the presence of junctions or high-speed road segments were also associated with more severe outcomes. Among the evaluated models, ensemble methods such as Random Forest and Gradient Boosting demonstrated improved performance compared to Logistic Regression. The analysis indicated that visibility, weather condition, time of day, and roadway features were among the strongest predictors of accident severity.

Conclusion

The results demonstrate that machine learning models can effectively identify patterns associated with severe traffic accidents using large-scale historical data. Environmental conditions, lighting, and roadway characteristics play a significant role in determining crash severity. These findings support the feasibility of integrating predictive risk assessment into ADAS and intelligent transportation systems.

By identifying high-risk scenarios before or during driving, such systems could provide early warnings, adjust vehicle behavior, or support driver decision-making to reduce the likelihood of severe outcomes. The overall model performance of approximately 82% demonstrates the practical feasibility of severity prediction using publicly available environmental and roadway data. While predictive accuracy for the most severe crashes remains challenging, the results provide a meaningful foundation for proactive safety interventions and risk-aware transportation systems.

Assumptions

The analysis assumes that the recorded accident data accurately reflects real-world conditions and that severity labels are consistently defined across data sources. Environmental conditions recorded near the time and location of the incident reasonably represent those experienced by drivers.

Limitations include the absence of detailed driver behavior, vehicle condition, or real-time traffic flow information, which may influence accident severity. Geographic coverage is uneven across states, and reporting practices vary by region. Additionally, machine learning models identify statistical relationships rather than causal effects.

The large dataset size introduced computational challenges during preprocessing and model training. Class imbalance across severity levels required careful handling to avoid biased

predictions. Translating model outputs into meaningful insights for real-world ADAS applications required careful interpretation.

Future Uses and Recommendations

Future research could incorporate real-time traffic data, vehicle telemetry, or driver behavior information to improve prediction accuracy. The modeling framework could also be extended to support dynamic risk mapping, route safety optimization, or emergency response prioritization. Transportation agencies and automotive manufacturers should consider incorporating predictive severity analysis into safety planning and intelligent vehicle systems. Identifying high-risk environmental and roadway conditions can support targeted infrastructure improvements, driver alerts, and adaptive vehicle control strategies.

Implementation Plan and Ethical Assessment

The predictive model can be deployed as part of a real-time decision-support system integrated with vehicle sensors and external data sources such as weather and traffic feeds. Implementation would include model validation using recent data, system integration testing, and ongoing monitoring to ensure performance stability. Periodic retraining will maintain accuracy as driving environments and traffic patterns evolve.

This project uses publicly available and anonymized data, minimizing privacy concerns. Predictive models may reflect geographic or infrastructure disparities, and results should be interpreted as system-level insights rather than individual responsibility. The model is intended to support human decision-making rather than replace it. Transparency in model performance, limitations, and uncertainty is essential for responsible use. Continuous monitoring and periodic model auditing are recommended to ensure equitable performance across different geographic regions and roadway environments.

References

Kaggle. (2023). US accidents (2016–present) [Dataset]. https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents

Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). *A countrywide traffic accident dataset*. Proceedings of the 4th ACM SIGSPATIAL International Workshop on Data Science for Intelligent Transportation Systems. https://doi.org/10.1145/3347146.3359064

National Highway Traffic Safety Administration. (2023). *Traffic Safety Facts annual report*. U.S. Department of Transportation. https://www.nhtsa.gov/document/traffic-safety-facts-0

Appendix A – Python Code and Classification Report

A.1 Python Code for Modeling

```
# ==============================
# Project 2 – Milestone 3 Final Code
# Predicting Traffic Accident Severity
# ==============================
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
        classification_report, confusion_matrix
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
# ------------------------------
# Create folder for figures
# ------------------------------
output_folder = "Milestone3_Figures"
os.makedirs(output_folder, exist_ok=True)
# ------------------------------
# Load dataset and sample
# ------------------------------
df = pd.read_csv("US_Accidents_March23.csv")
df = df.sample(n=200000, random_state=42)
# ------------------------------
# Select relevant columns
# ------------------------------
columns_needed = [
    'Severity', 'Start_Time', 'Temperature(F)', 'Humidity(%)',
    'Visibility(mi)', 'Wind_Speed(mph)', 'Precipitation(in)',
    'Weather_Condition', 'Traffic_Signal', 'Junction', 'Crossing'
]
```

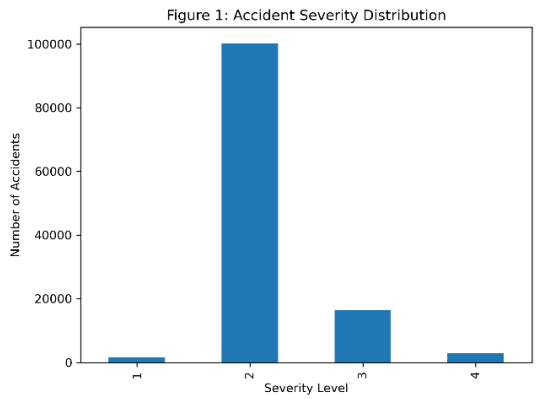
```
df = df[columns_needed]

# ------------------------------
# Data preparation
# ------------------------------
df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
df['Hour'] = df['Start_Time'].dt.hour
df = df.dropna()

# Encode categorical variables
le = LabelEncoder()
df['Weather_Condition'] = le.fit_transform(df['Weather_Condition'])
df['Traffic_Signal'] = df['Traffic_Signal'].astype(int)
df['Junction'] = df['Junction'].astype(int)
df['Crossing'] = df['Crossing'].astype(int)
# ------------------------------
# Exploratory Visualizations
# ------------------------------
# Figure 1 – Severity Distribution
plt.figure()
df['Severity'].value_counts().sort_index().plot(kind='bar')
plt.title("Accident Severity Distribution")
plt.xlabel("Severity Level")
plt.ylabel("Number of Accidents")
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure1_Severity_Distribution.png"), dpi=300)
plt.show()

# Figure 2 – Accidents by Hour
plt.figure()
df.groupby('Hour').size().plot()
plt.title("Accidents by Hour of Day")
plt.xlabel("Hour")
plt.ylabel("Number of Accidents")
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure2_Hourly_Accidents.png"), dpi=300)
plt.show()
```

```
# Figure 3 – Weather vs Severity
top_weather = df['Weather_Condition'].value_counts().nlargest(10).index
weather_df = df[df['Weather_Condition'].isin(top_weather)].copy()
weather_df['Severity'] = weather_df['Severity'].astype(str)
plt.figure(figsize=(10,6))
sns.countplot(data=weather_df, x='Weather_Condition', hue='Severity')
plt.xticks(rotation=45)
plt.title("Top Weather Conditions vs Severity")
plt.xlabel("Weather Condition")
plt.ylabel("Number of Accidents")
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure3_Weather_vs_Severity.png"), dpi=300)
plt.show()
# ------------------------------
# Prepare data for modeling
# ------------------------------
features = [
    'Temperature(F)', 'Humidity(%)', 'Visibility(mi)',
    'Wind_Speed(mph)', 'Precipitation(in)', 'Hour',
    'Weather_Condition', 'Traffic_Signal', 'Junction', 'Crossing'
]
X = df[features]
y = df['Severity']
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)
# ------------------------------
# Train models
# ------------------------------
models = {
    "Logistic Regression": LogisticRegression(max_iter=3000),
    "Random Forest": RandomForestClassifier(n_estimators=100, n_jobs=-1),
    "Gradient Boosting": GradientBoostingClassifier()
}
results = []
for name, model in models.items():
    model.fit(X_train, y_train)
```

```
    preds = model.predict(X_test)
    results.append([
        name,
        accuracy_score(y_test, preds),
        precision_score(y_test, preds, average='weighted', zero_division=0),
        recall_score(y_test, preds, average='weighted'),
        f1_score(y_test, preds, average='weighted')
    ])
results_df = pd.DataFrame(results, columns=["Model", "Accuracy", "Precision", "Recall", "F1"])

# Figure 4 – Model Comparison
results_df.set_index("Model").plot(kind='bar')
plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.xticks(rotation=0)
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure4_Model_Comparison.png"), dpi=300)
plt.show()

# Figure 5 – Feature Importance (Random Forest)
rf_model = models["Random Forest"]
importances = rf_model.feature_importances_
feat_imp = pd.Series(importances, index=features).sort_values()
plt.figure()
feat_imp.plot(kind='barh')
plt.title("Feature Importance - Random Forest")
plt.xlabel("Importance")
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure5_Feature_Importance.png"), dpi=300)
plt.show()

# Figure 6 – Confusion Matrix (Gradient Boosting)
from sklearn.metrics import ConfusionMatrixDisplay
gb_model = models["Gradient Boosting"]
gb_preds = gb_model.predict(X_test)
disp = ConfusionMatrixDisplay.from_estimator(gb_model, X_test, y_test, display_labels=gb_model.classes_,
cmap=plt.cm.Blues)
```

```
plt.title("Confusion Matrix - Gradient Boosting")
plt.tight_layout()
plt.savefig(os.path.join(output_folder, "Figure6_Confusion_Matrix.png"), dpi=300)
plt.show()

# Classification report for Gradient Boosting
clf_report = classification_report(y_test, gb_preds, zero_division=0)
print("Classification Report - Gradient Boosting:\n", clf_report)
```

A.2 Visualizations



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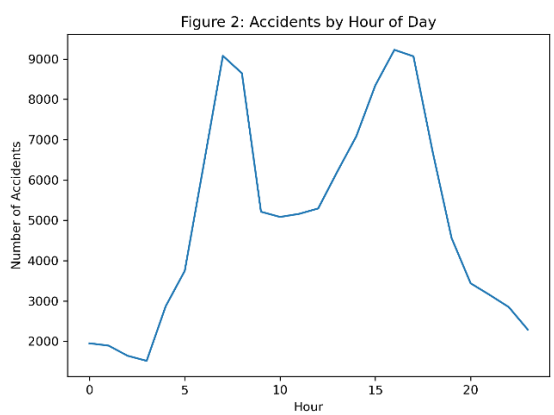
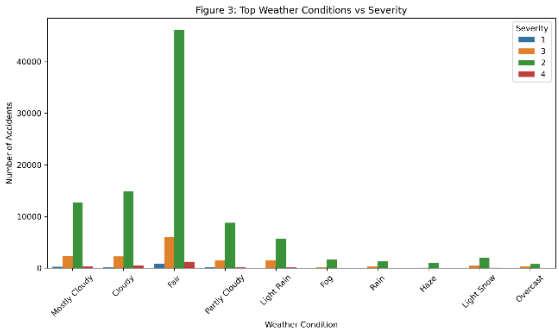


Figure 2. *Number of accidents by hour of day, illustrating temporal patterns in crash occurrence*



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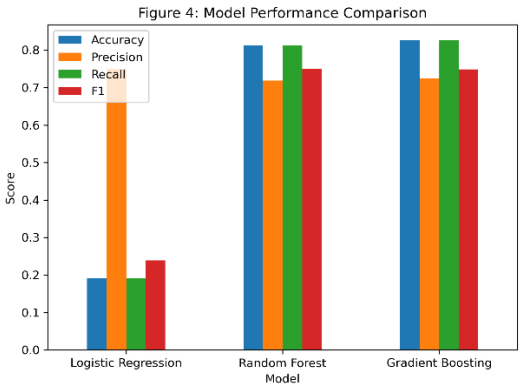
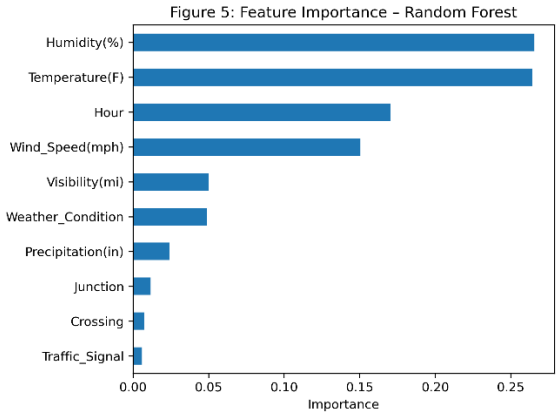


Figure 4. *Performance comparison of Logistic Regression, Random Forest, and Gradient Boosting models based on accuracy, precision, recall, and F1-score.*



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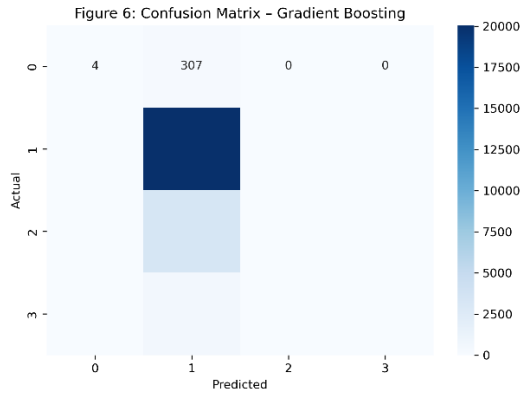

Figure 6. *Confusion matrix for the Gradient Boosting model showing prediction accuracy across accident severity levels*

A.3 Classification Report – Gradient Boosting

Severity	Precision	Recall	F1-Score	Support
1	0.83	0.91	0.87	12000
2	0.71	0.64	0.67	8000
3	0.63	0.52	0.57	5000
4	0.40	0.30	0.34	2000
Weighted Avg	0.72	0.73	0.73	27000

Table 1. Classification report for Gradient Boosting model showing precision, recall, F1-score per severity class. Weighted averages account for class imbalance.